

Phase 1: Foundations of Generative AI

1. What is Generative AI?
 2. How is Generative AI different from traditional AI?
 3. What are some real-world applications of Generative AI?
 4. What is a Large Language Model (LLM)?
 5. What are the major stages in the LLM lifecycle?
 6. What is pre-training in an LLM?
 7. What is fine-tuning in an LLM?
 8. What happens during inference?
 9. What is a token in LLMs?
 10. Why are tokens important for language models?
 11. What are embeddings?
 12. Why are embeddings used in AI systems?
 13. What is the Transformer architecture?
 14. What problem does the attention mechanism solve?
 15. Name some popular LLM providers and models.
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Phase 2: Prompt Engineering

16. What is prompt engineering?
 17. Why is prompt engineering important?
 18. What is a system prompt?
 19. What is a user prompt?
 20. What is the difference between a system prompt and a user prompt?
 21. What is zero-shot prompting?
 22. What is few-shot prompting?
 23. What is Chain-of-Thought prompting?
 24. When would you use few-shot prompting instead of zero-shot prompting?
 25. What are some common prompt optimization techniques?
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Phase 3: Embeddings and Vector Databases

26. What is an embedding vector?
27. How are embeddings generated?
28. Why are embeddings useful for semantic search?
29. What is similarity search?
30. What is cosine similarity?
31. What is the difference between keyword search and semantic search?
32. What is a vector database?
33. Why can't traditional relational databases efficiently handle vector search?
34. What is FAISS?

- 35. What is Pinecone?
 - 36. What is Weaviate?
 - 37. What is chunking in RAG systems?
 - 38. Why is chunk size important?
 - 39. What challenges arise from very large chunks?
 - 40. What challenges arise from very small chunks?
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Phase 4: Retrieval-Augmented Generation (RAG)

- 41. What is RAG?
 - 42. Why do we need RAG when we already have LLMs?
 - 43. What are the main components of a RAG pipeline?
 - 44. What is document retrieval?
 - 45. What is document indexing?
 - 46. What is hybrid search?
 - 47. What is reranking in RAG systems?
 - 48. How does RAG help reduce hallucinations?
 - 49. What is context injection in RAG?
 - 50. What is the difference between chunking and windowing?
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Phase 5: Agents and Tool Calling

- 51. What is an AI agent?
 - 52. How is an AI agent different from a chatbot?
 - 53. What is tool calling?
 - 54. Why do agents need external tools?
 - 55. What is function calling in LLMs?
 - 56. What is agent memory?
 - 57. What is short-term memory in agents?
 - 58. What is long-term memory in agents?
 - 59. What is a multi-agent system?
 - 60. What are the advantages of multi-agent systems?
 - 61. What is LangChain?
 - 62. What is AutoGen?
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Phase 6: Fine-Tuning and Customization

- 63. What is fine-tuning?
- 64. When should you choose fine-tuning over prompt engineering?
- 65. What is LoRA?

66. What is QLoRA?
67. Why is LoRA more efficient than full fine-tuning?
68. What factors should be considered while preparing a fine-tuning dataset?
69. How do you evaluate a fine-tuned model?
70. What metrics can be used to measure model performance?

what is Vectorless RAG?

What is a corrective argument?

What is a self-RAG?

What is RAG Fusion?